CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 3f: Joint Distributions

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Outline

1. Review
2. Intro: Joint Distributions
3. Discrete random variables
4. Continuous random variables

Midterm Exam

- Open-resource examination
- Available for download via Moodle on Tuesday, October 20th at **12:50 PM**
- Due **same day** at **2:30 PM** on Gradescope
- Exam is designed to be completed in under **75 minutes**. The 100-minute window gives you flexibility and time to plan, organize and check your work before submission.

Set theory

Properties:

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$
$$(A \cup B) \cup C = A \cup (B \cup C)$$
$$(AB)C = A(BC)$$
$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$
$$(AB) \cup C = (A \cup C) \cap (B \cup C)$$

De Morgan's rule:

$$\overline{E_1 \cup E_2 \cup \cdots \cup E_n} = \overline{E_1} \cap \overline{E_2} \cap \cdots \cap \overline{E_n}$$
$$\overline{E_1 \cap E_2 \cap \cdots \cap E_n} = \overline{E_1} \cup \overline{E_2} \cup \cdots \cup \overline{E_n}$$

Probability theory

General rules:

$$P(\overline{E}) = 1 - P(E)$$

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

$$P(A \cap B) = P(AB) = P(A|B)P(B)$$

$$P(A|B) = \frac{P(AB)}{P(B)}$$

$$P(A|B) = 1 - P(\overline{A}|B)$$

Mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

Statistically independent events:

$$P(AB) = P(A)P(B)$$

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

Total probability theorem:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + \cdots + P(A|E_n)P(E_n) = \sum_{i=1}^{n} P(A|E_i)P(E_i)$$

Bayes' theorem (general):

$$P(E_i|A) = \frac{P(A|E_i)P(E_i)}{\sum_{j=1}^{n} P(A|E_j)P(E_j)}$$

Bayes' theorem (special case for two events):

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|C)P(C)}$$

Counting

Number of ways to arrange n objects:

$$n! = n(n-1)(n-2)\cdots 3 \times 2 \times 1$$

Permutations (arrangements) of n objects: $n!$
Permutations (arrangements) of n objects taken k at a time:

$${}_nP_k = \frac{n!}{(n-k)!}$$

Permutations of n objects with n_1 of one type, n_2 of another type, $\ldots$, n_k of the k-th type:

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Combinations of n objects taken k at a time:

$${}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Probability distributions

Distribution	**Mean**	**Variance**	**Median**	CDF using `scipy.stats` $P(X \leq x)$
Uniform(a, b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{a+b}{2}$	`uniform.cdf(x, a, b-a)`
$\mathcal{N}(\mu, \sigma)$	μ	σ^2	μ	`norm.cdf(x, mu, sigma)`
$\mathcal{LN}(\mu, \sigma)$	$\exp\left(\mu + \frac{1}{2}\sigma^2\right)$	$\left[\exp\left(\sigma^2\right) - 1\right] \exp(2\mu + \sigma^2)$	$\exp\left(\mu\right)$	`lognorm.cdf(x, sigma, scale=np.exp(mu))`
Exponential(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\ln 2}{\lambda}$	`expon.cdf(x, scale=1/lambda)`
Binomial(n, p)	np	$np(1-p)$	$\lfloor (n+1)p \rfloor$	`binom.cdf(x, n, p)`
Poisson(λ)	λ	λ	$\lfloor \lambda \rfloor$	`poisson.cdf(x, lambda)`

Note: the $\lfloor \cdot \rfloor$ symbol means to round down to the nearest integer.

Today's objectives

Joint distributions:

- Understand the concept of jointly distributed random variables
- Differentiate between marginal and conditional distributions
- Manipulate joint distributions to compute probabilities

Joint distributions

Given two random variables X and Y:

Discrete case

The joint PMF is:

$$p_{X,Y}(x_i, y_j) = P(X = x_i, Y = y_j) \tag{1}$$

The CDF is:

$$F_{X,Y}(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p_{X,Y}(x_i, y_j) \tag{2}$$

Continuous case

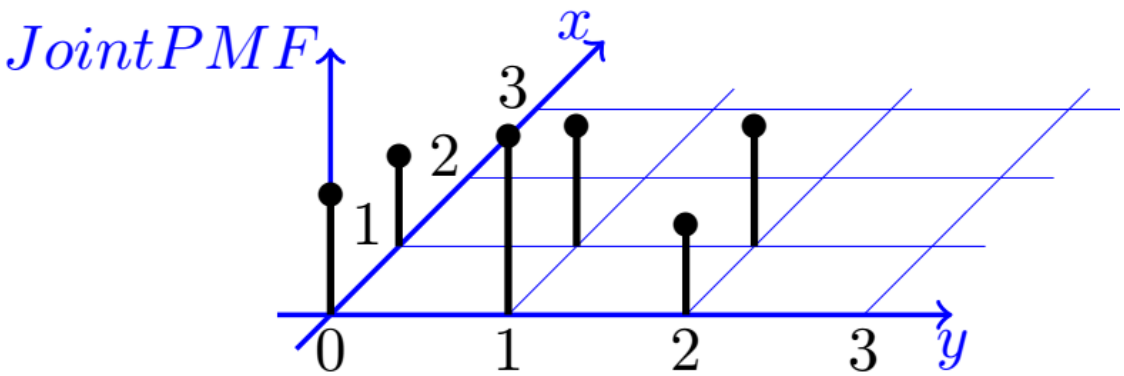
The joint probability is given by:

$$P(a < X \leq b, c < Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y) dy dx \tag{3}$$

Example 1: Joint PMF of two random variables

Given two random variables X and Y with joint PMF indicated in the table below:

	$Y=0$	$Y=1$	$Y=2$	$p_X(x)$
$X=0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$	?
$X=1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$	?
$p_Y(y)$	?	?	?	



(a) Find $P(X=0, Y \leq 1)$
(b) Find the marginal PMFs of X and Y
(c) Find $P(Y=1|X=0)$
(d) Are X and Y independent?

Example 1: Joint PMF of two random variables (cont.)

Solution

(a) To find the probability $P(X = 0, Y \leq 1)$, simply add up the cells *jointly* satisfying the conditions.

	$Y = 0$	$Y = 1$	$Y = 2$	$p_X(x)$
$X = 0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$	?
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$	?
$p_Y(y)$	?	?	?	

$$\begin{aligned} P(X = 0, Y \leq 1) &= \frac{1}{6} + \frac{1}{4} \\ &= \frac{4+6}{24} = \frac{10}{24} = \frac{5}{12} \end{aligned}$$

Example 1: Joint PMF of two random variables (cont.)

Solution

(b) To find the marginal PMFs of X and Y, i.e. $p_X(x) = P(X = x)$ and $p_Y(y) = P(Y = y)$, respectively, sum up the rows and columns in the table:

	$Y = 0$	$Y = 1$	$Y = 2$	$p_X(x)$
$X = 0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{6} + \frac{1}{4} + \frac{1}{8} = \frac{13}{24}$
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{8} + \frac{1}{6} + \frac{1}{6} = \frac{11}{24}$
$p_Y(y)$	$\frac{1}{6} + \frac{1}{8} = \frac{7}{24}$	$\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$	$\frac{1}{8} + \frac{1}{6} = \frac{7}{24}$	

Thus, we obtain:

$$p_X(x) = \begin{cases} \frac{13}{24}, & x = 0 \\ \frac{11}{24}, & x = 1 \\ 0, & \text{otherwise} \end{cases} \qquad p_Y(y) = \begin{cases} \frac{7}{24}, & y = 0 \\ \frac{5}{12}, & y = 1 \\ \frac{7}{24}, & y = 2 \\ 0, & \text{otherwise} \end{cases}$$

Example 1: Joint PMF of two random variables (cont.)

	$Y=0$	$Y=1$	$Y=2$	$p_X(x)$
$X=0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{6}+\frac{1}{4}+\frac{1}{8}=\frac{13}{24}$
$X=1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{8}+\frac{1}{6}+\frac{1}{6}=\frac{11}{24}$
$p_Y(y)$	$\frac{1}{6}+\frac{1}{8}=\frac{7}{24}$	$\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$	$\frac{1}{8}+\frac{1}{6}=\frac{7}{24}$	

Solution

(c) Here we use the conditional probability formula:

$$P(Y=1|X=0) = \frac{P(Y=1\cap X=0)}{P(X=0)} = \frac{P(Y=1,X=0)}{P(X=0)}$$

$$= \frac{p_{XY}(0,1)}{p_X(0)} = \frac{\frac{1}{4}}{\frac{13}{24}} = \frac{6}{13}$$

Example 1: Joint PMF of two random variables (cont.)

	$Y=0$	$Y=1$	$Y=2$	$p_X(x)$
$X=0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{6}+\frac{1}{4}+\frac{1}{8}=\frac{13}{24}$
$X=1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{8}+\frac{1}{6}+\frac{1}{6}=\frac{11}{24}$
$p_Y(y)$	$\frac{1}{6}+\frac{1}{8}=\frac{7}{24}$	$\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$	$\frac{1}{8}+\frac{1}{6}=\frac{7}{24}$	

Solution

(d) Two r.v.'s are independent if $P(Y=y_j|X=x_i)=P(X=x_i)$ or $P(X=x_i|Y=y_j)=P(Y=y_j)$ for all i and j. Here,

$$P(Y=1|X=0)=\frac{6}{13}\neq P(Y=1)=\frac{5}{12}$$

The independence condition fails. Hence X and Y are not independent.

Conditional distributions of continuous random variables

Recall the definition of conditional probability (multiplication rule):

$$P(A|B) = \frac{P(AB)}{P(B)} \tag{4}$$

Similarly, for two continuous r.v.'s, the conditional PDF of X given Y is:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \tag{5}$$

Joint PDF and CDF of two variables

The joint PDF is given by:

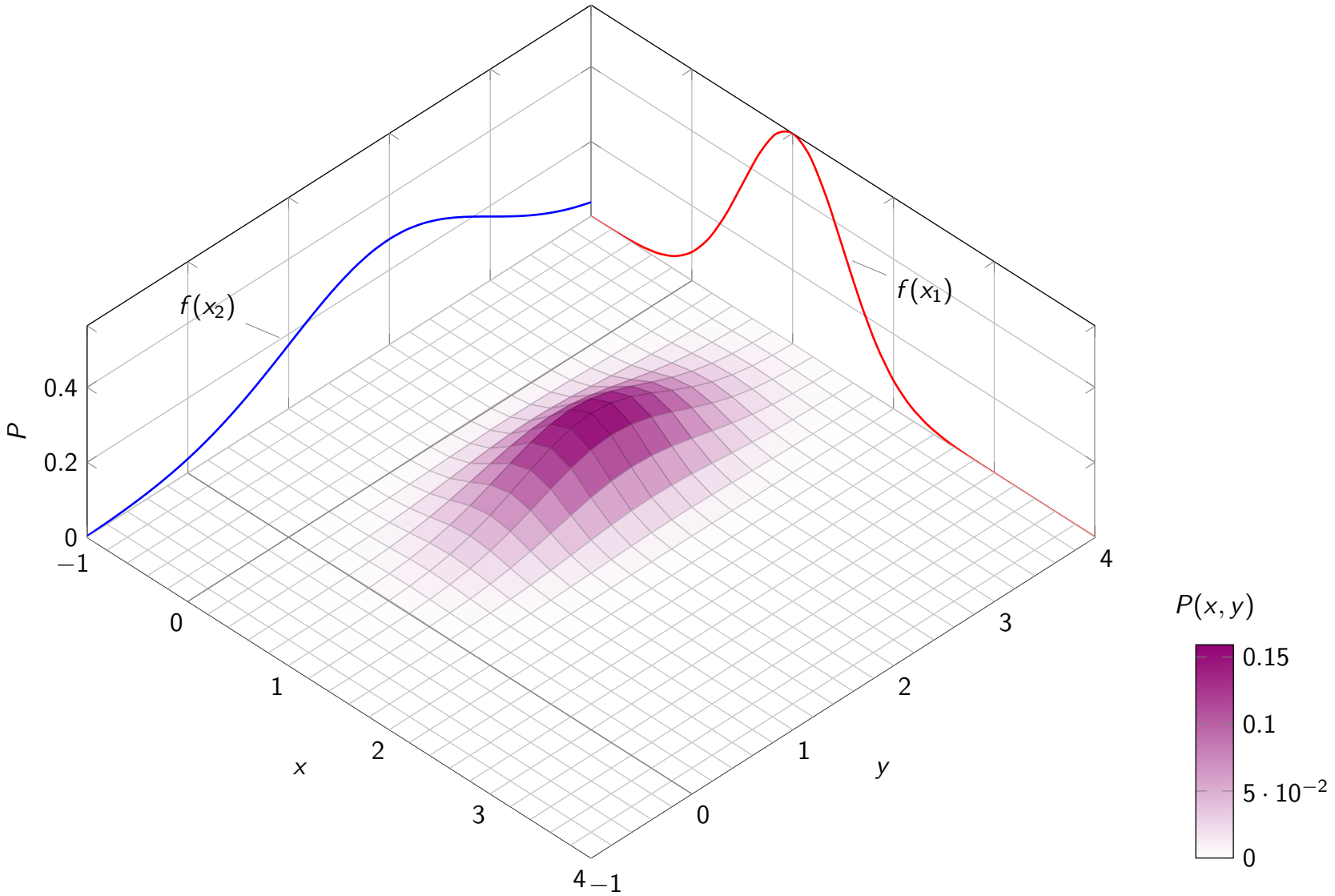
$$f_{X,Y}(x,y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x) \tag{6}$$

While the joint CDF is given by:

$$F_{X,Y}(a,b) = P(X \leq a, Y \leq b) = \int_{-\infty}^{a}\int_{-\infty}^{b} f_{X,Y}(x,y)dydx \tag{7}$$

Joint and marginal distributions

Marginal distributions: $f(x)$ and $f(y)$
Joint distribution: $f(x, y)$.



Marginal distributions of continuous random variables

Recall the theorem of total probability:

$$P(A) = \sum_{i=1}^{n} P(A|E_i)P(E_i) \tag{8}$$

Similarly, the marginal PDFs from a joint distribution of two continuous r.v.'s X and Y is given as:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y)dy = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy \tag{9}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y|x)f_X(x)dx = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx \tag{10}$$

Example 2: Water levels

The daily water levels of two reservoirs A and B are denoted by two r.v.'s X and Y having the following joint PDF:

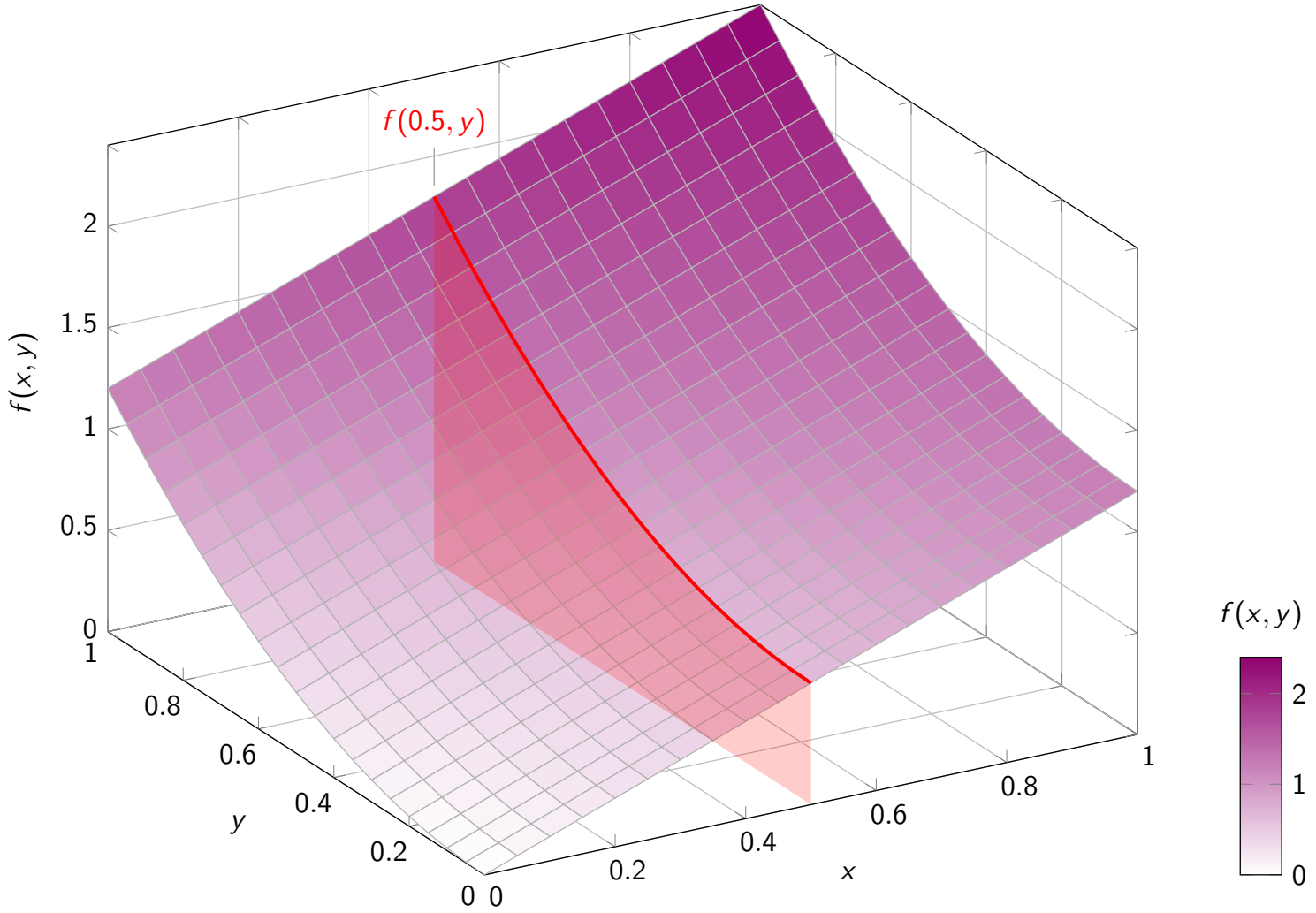
$$f(x,y) = \frac{6}{5}\left(x + y^2\right), \quad 0 < x < 1; 0 < y < 1$$

(a) Determine the marginal density function of the daily water level for reservoir A, i.e. $f_X(x)$

(b) If reservoir A is half full on a given day, what is the probability that the water level will be more than half full?

Example 2: Water levels (cont.)

$$f(x,y) = \frac{6}{5}\left(x + y^2\right), \quad 0 < x < 1; 0 < y < 1$$



Example 2: Water levels (cont.)

$$f(x,y) = \frac{6}{5}\left(x + y^2\right), \quad 0 < x < 1; 0 < y < 1$$

Solution

(a) We obtain the marginal density function $f_X(x)$ by integrating the joint PDF over y:

$$\begin{aligned} f_X(x) &= \int_0^1 \frac{6}{5}\left(x + y^2\right) dy \\ &= \frac{6}{5}\left[xy + \frac{y^3}{3}\right]_0^1 \\ &= \frac{2}{5}(3x+1) \quad (0 < x < 1) \end{aligned}$$

This is the **marginal distribution**

Example 2: Water levels (cont.)

$$f(x,y) = \frac{6}{5}\left(x + y^2\right), \quad 0 < x < 1; 0 < y < 1$$

Solution

(b) If reservoir A is half full on a given day, what is the probability that the water level will be more than half full? We first find the conditional distribution:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\frac{6}{5}\left(x+y^2\right)}{\frac{2}{5}\left(3x+1\right)} = 3\frac{x+y^2}{3x+1}$$

$$\text{Thus: } P(Y > 0.5|X = 0.5) = \int_{0.5}^{1} f_{Y|X}(y|x = 0.5)dy$$

$$= 3\int_{0.5}^{1} \frac{0.5+y^2}{1.5+1}dy$$

$$= \left(\frac{3}{2.5}\right)\left[0.5y + \frac{y^3}{3}\right]_{0.5}^{1} = \boxed{0.65}$$